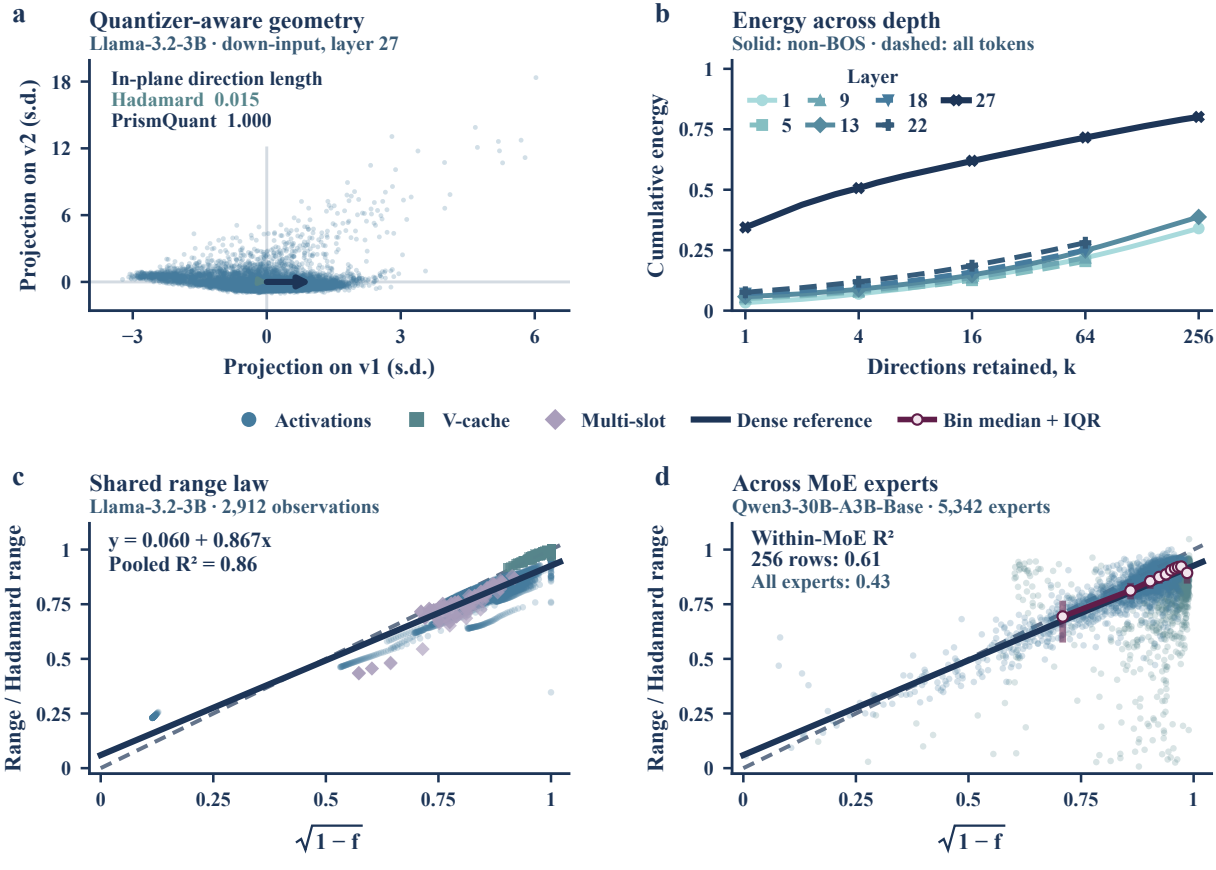

a
Quantizer-aware geometry
Llama-3.2-3B · down-input, layer 27
In-plane direction length
Hadamard 0.015
PrismQuant 1.000
Projection on v2 (s.d.)
Projection on v1 (s.d.)
b
Energy across depth
Solid: non-BOS · dashed: all tokens
Layer
1
5
9
13
18
22
27
Cumulative energy
Directions retained, k
Activations
V-cache
Multi-slot
Dense reference
Bin median + IQR
c
Shared range law
Llama-3.2-3B · 2,912 observations
y = 0.060 + 0.867x
Pooled R² = 0.86
Range / Hadamard range
√1 − f
d
Across MoE experts
Qwen3-30B-A3B-Base · 5,342 experts
Within-MoE R²
256 rows: 0.61
All experts: 0.43
Range / Hadamard range
√1 − f